

Report

Nara Avila Moraes

Physics Institute
IFUSP

2026 / 11 / 07

Summary

1. Tasks Status
2. Synthetic Data Studies
3. Selected Article

Conference Materials Status

Task	Status	
2026-10 Conference	✓	Simpósio IFUSP Presentation
2026-10 Conference	✓	AGORÁ Presentation
2026-11 Poster	◕	3rd Schooll QC Poster
2026-12 Conference	○	LASNPA Presentation
2026-12 Paper	○	LASNPA Proceedings
2027-03 Conference	○	Quark Matter Presentation

Study Phases – Dependencies

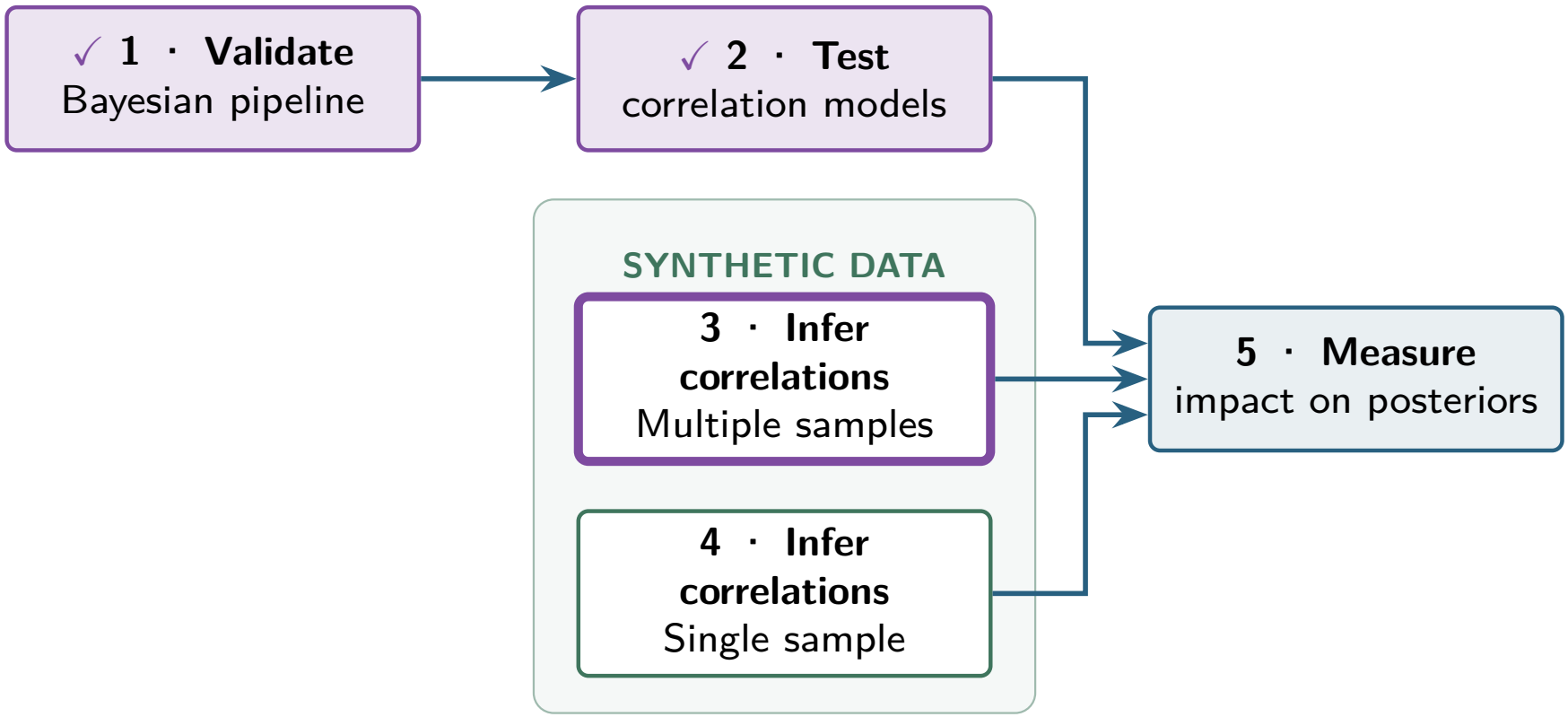


Figure 1: Dependencies between the study phases.

Phase 1 – Bayesian Pipeline Validation

Task	**Status**	**Notebook**
↪ Reproduce Mankolli et al. results with the original JETSCAPE emulator	✓	Notebook
↪ Reproduce the same results with the `hic_bayes` emulator	✓	Notebook
↪ Emulator closure test	✓	Notebook
↪ MCMC convergence test	✓	Notebook

Phase 2 – Correlation Model Sensitivity

Task	Status	Output
↪ Au–Au-200 + d–Au-200: Correlation-model sensitivity		Posteriors
• Global CS correlation	✓	Notebook
• Global RBF correlation	✓	Notebook
• Global RBF correlation fix unit	✓	Notebook
• Sector CS correlation	✓	Notebook
• Sector RBF correlation	✓	Notebook
↪ Compare posterior shifts, credible intervals, and parameter correlations		Analysis
• Global correlation models	✓	Notebook
• Detector-grouped correlation models	✓	Notebook
↪ Quantify and compare the emulator-error and experimental-error covariance matrices in the likelihood	◑	Analysis

Phase 3 – Correlation Inference from Multiple Samples

Task	**Status**	**Output**
↪ Formulate questions about the study	◑	Questions
↪ Generate controlled synthetic datasets with known covariance (N samples; goal: $N = 10$)	✓	Dataset
↪ Classical correlation inference	✓	Benchmark
↪ Quantum-learning correlation inference	✓	Benchmark
↪ Classical vs. quantum comparison against the known covariance	✓	Comparison

Phase 4 – Single-Sample Correlation Inference

Task	**Status**	**Output**
↪ Study correlation approximations when only one experimental sample is available	◔	Heuristics
↪ Use reported uncertainty components to constrain plausible correlation structures	○	Models

Phase 5 – End-to-End Bayesian Impact

Task	Status	Output
↪ Propagate inferred covariance matrices through the Bayesian likelihood		Posteriors
• **Single sample:** CS, RBF, Zero, GP	◕	
• **Multiple samples:** Classical, Quantum	○	
↪ Quantify covariance-estimation error → posterior error	○	Final study

Synthetic Data Studies

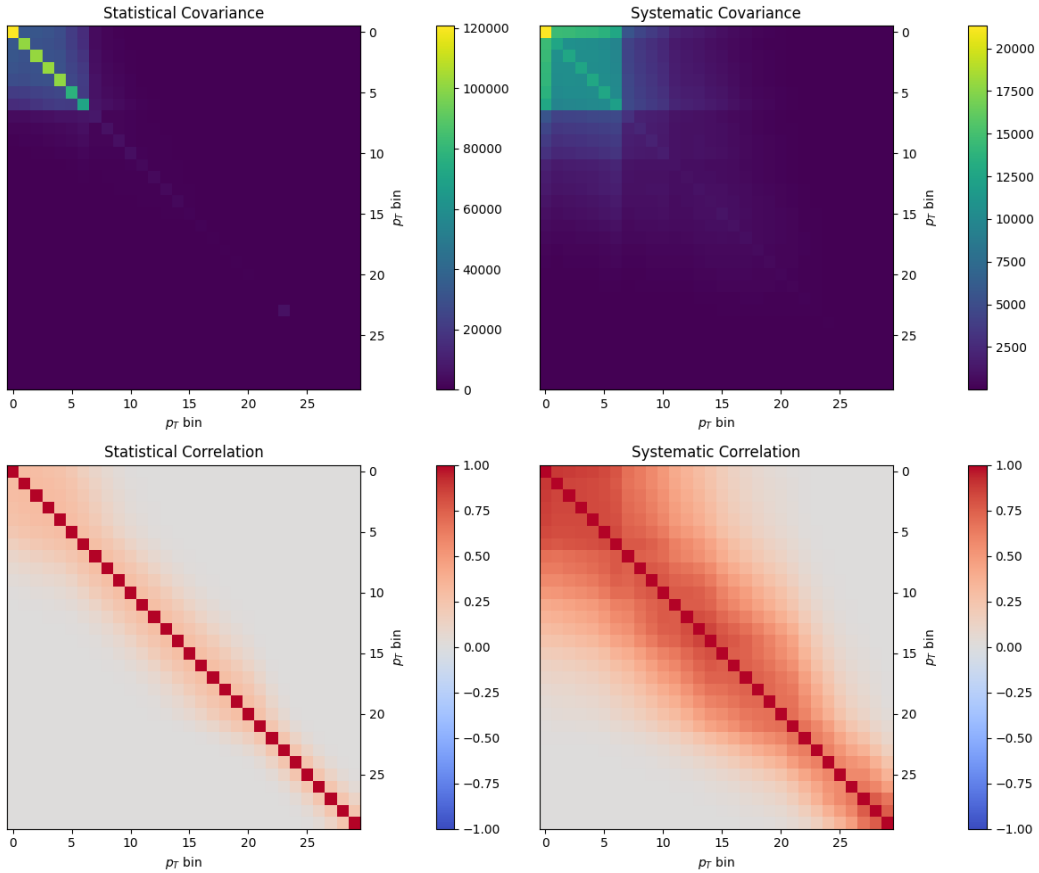


Figure 2: Statistical and systematic covariance used to construct the synthetic data.

Synthetic Data Studies

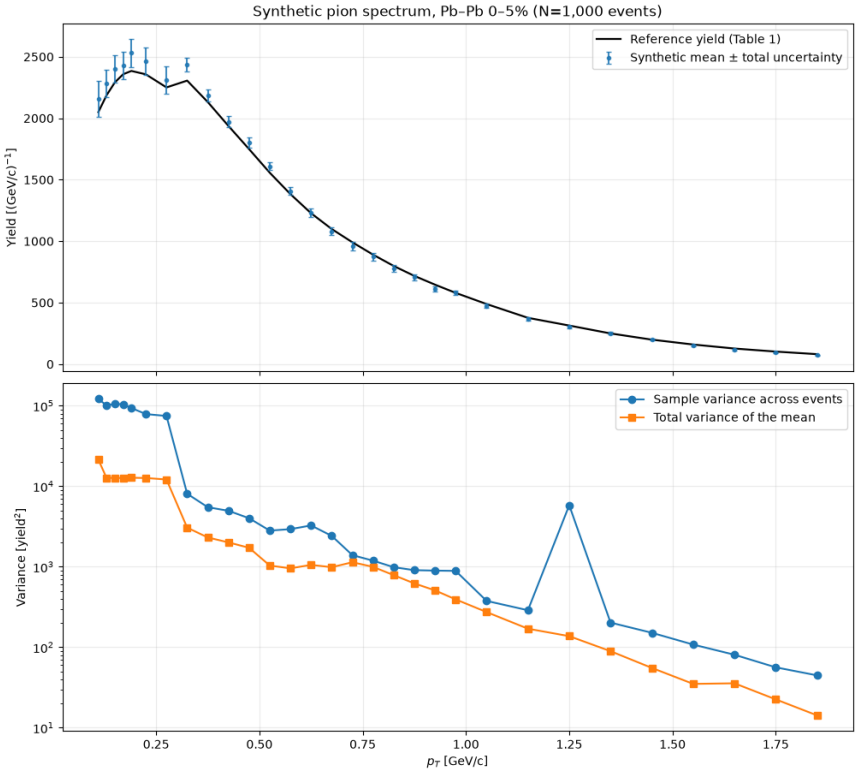


Figure 3: Synthetic bin means and variances, using ALICE pion spectra in 0–5% central Pb–Pb collisions at $\sqrt{s_{\mathrm{NN}}} = 5.02$ TeV as reference to generate event-by-event data with assumed statistical and systematic covariance structures, as shown on slide 10 (HEPData ins1759506, v1, Table 1).

Classical vs Quantum correlation estimation

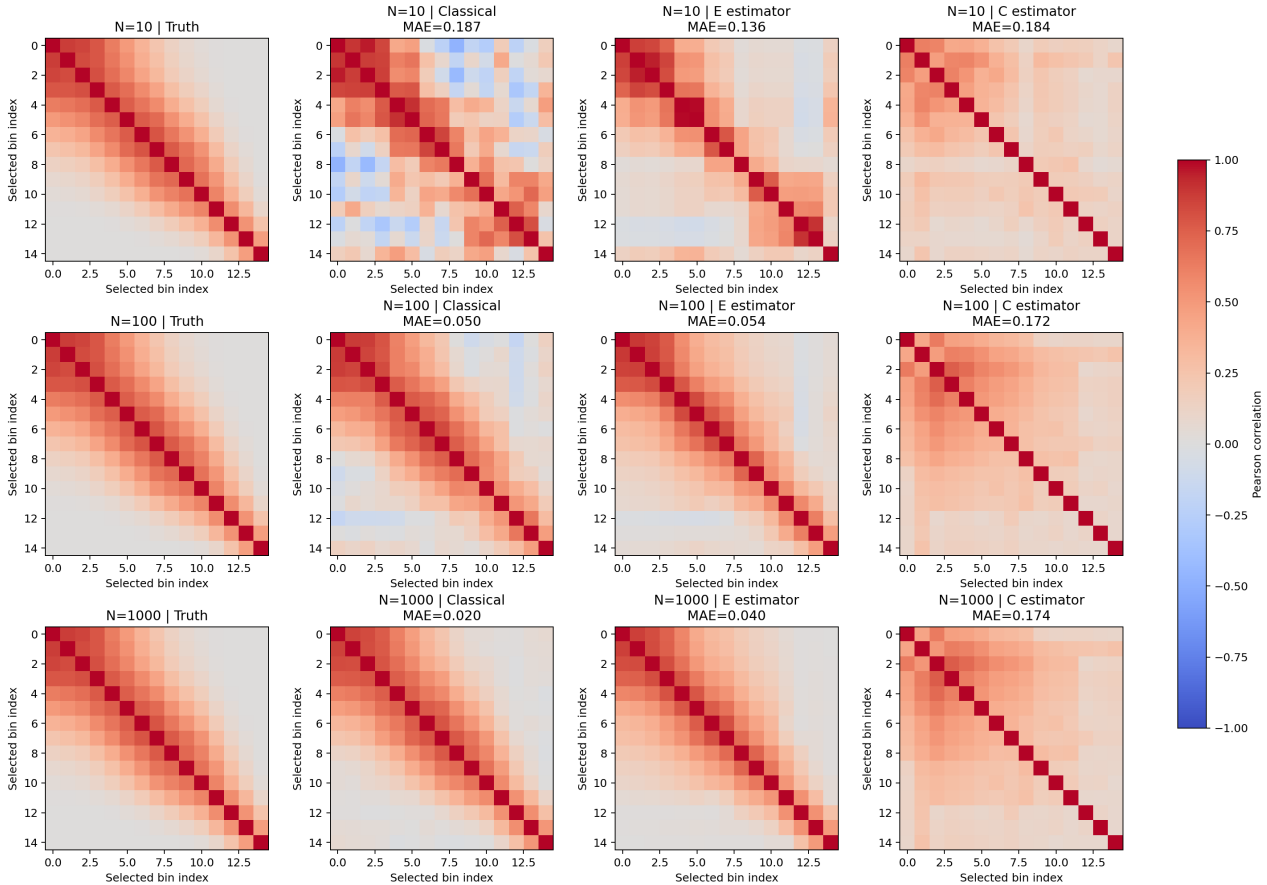


Figure 4: Comparing classical and quantum correlation learning using 10, 100, and 1000 samples. Only 15 bins were used due to the computational limitations of the quantum circuit simulator.

Selected Article

Unexpectedly large violation of isospin symmetry in nucleus–nucleus collisions at CERN SPS energies

Andrzej Rybicki[1,*]
for the NA61/SHINE Collaboration

[1]The Henryk Niewodniczański Institute of Nuclear Physics Polish Academy of Sciences, ul. Radzikowskiego 152, 31-342 Kraków, Poland

Abstract. The NA61/SHINE experiment at the CERN SPS reports an unexpectedly large excess in the inclusive production of charged over neutral kaons in 0–10% central Ar+Sc collisions at $\sqrt{s_{\mathrm{NN}}} = 8.8$ and 11.9 GeV. This effect corresponds to an unexpectedly large violation of isospin symmetry, that cannot be explained by known mechanisms contained in the present theoretical models. The comparison of the NA61/SHINE results to data from other experiments reveals, within sizeable uncertainties, a consistent isospin symmetry-violating excess of charged kaon production in the energy regime $2.6 < \sqrt{s_{\mathrm{NN}}} < 200$ GeV. At LHC energies ($0.9 < \sqrt{s_{\mathrm{NN}}} < 13$ TeV) the charged-over-neutral kaon ratio appears consistent with unity, but the uncertainties remain sizeable. A similar excess of charged over neutral kaon yields is observed by NA61/SHINE in minimum bias π^-+C reactions at $\sqrt{s_{\mathrm{NN}}} = 17.3$ and 25.7 GeV, also in disagreement with model predictions. No clear theoretical explanation of these phenomena has been provided so far. An unexpectedly large violation of isospin symmetry brings potential heavy consequences for our understanding of particle production, and calls for a concentrated experimental and theoretical effort.

Figure 5: Isospin symmetry violation in nucleus–nucleus collisions.